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WO-100-02 — Bench test relief valve 3"-P150-RV-4002/2 on T-101-N3

Area	A100 Tank farm	**Task**	PR Relief valve bench test and reinstatement
Priority	1 (safety)	**Issued by**	Facility owner (demonstration)
Primary asset	T-101 — T-101 tank 12' dia × 22' high — E 25'-0" N 26'-6" EL 100'-6"		
Equipment	T-101	**Lines**	3"-P150-RV-4002, 4"-P150-FL-4005
Isometrics	ISO-RV-4002, ISO-FL-4005 (owner's drawings, attached)		
Permits	General work permit, Lifting plan		

Reason

Relief valve due for its periodic bench test.

Scope of work

1. Confirm T-101 depressured and isolated.
2. Unbolt and remove VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET (3"-P150-RV-4002/2); blind the nozzle and the outlet line.
3. Bench test: set pressure 150 psi and seat tightness (SW-TEST-3); service the valve as found.
4. Reinstall with new gaskets and studs, tightened per SW-WO-5; remove blinds.

Affected items: 3"-P150-RV-4002/2, 3"-P150-RV-4002/1, 4"-P150-FL-4005/A1

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-3, SW-WO-1, SW-WO-2, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-RV-4002	3"-P150-RV-4002	3"-P150-RV-4002	1	agrees with the model
ISO-FL-4005	4"-P150-FL-4005	4"-P150-FL-4005	3	agrees with the model

ISO-RV-4002 (PDF): ../isometrics-imported/ISO-RV-4002.pdf · ISO-FL-4005 (PDF): ../isometrics-imported/ISO-FL-4005.pdf — the sheets that hold the work follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
T-101	equipment	T-101 tank 12' dia × 22' high		E 25'-0" N 26'-6" EL 100'-6"
3"-P150-RV-4002/2	item	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 124'-0"
3"-P150-RV-4002/1	item	GASKET 3" P150 COMPRESSED SHEET	3"-P150-RV-4002	E 22'-0" N 26'-6" EL 123'-4-3/8"
4"-P150-FL-4005/A1	item	GASKET 4" P150 COMPRESSED SHEET	4"-P150-FL-4005	E 22'-0" N 25'-10-1/16" EL 124'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	T-101	3"-P150-RV-4002	T-101 depressured, drained and isolated under the shutdown isolation (no inlet valve)			
2	blind	3"-P150-RV-4002/1	3"-P150-RV-4002	Blind the nozzle T-101-N3 once the valve is removed	E 22'-0" N 26'-6" EL 123'-4-5/16"		
3	blind	4"-P150-FL-4005/A1	4"-P150-FL-4005	Blind the outlet line 4"-P150-FL-4005 at its first flange	E 22'-0" N 25'-10-1/8" EL 124'-0"		

Parts

Description	Qty	Replaces
GASKET 3" P150 COMPRESSED SHEET	1	3"-P150-RV-4002/1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150	1	3"-P150-RV-4002/3
GASKET 4" P150 COMPRESSED SHEET	1	4"-P150-FL-4005/A1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	1	4"-P150-FL-4005/A3

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
3"-P150-RV-4002/1	4 × 0.875" studs (3"-P150-RV-4002/3)	27,059	3	
4"-P150-FL-4005/A1	4 × 1" studs (4"-P150-FL-4005/A3)	35,343	3	

Test — Relief valve bench test

Description	Set pressure 150 psi (class design pressure, PSV-1) and seat tightness
Pressure	150 psi
Basis	SW-TEST-3

Inspection and test plan (SW-ITP-1..3)

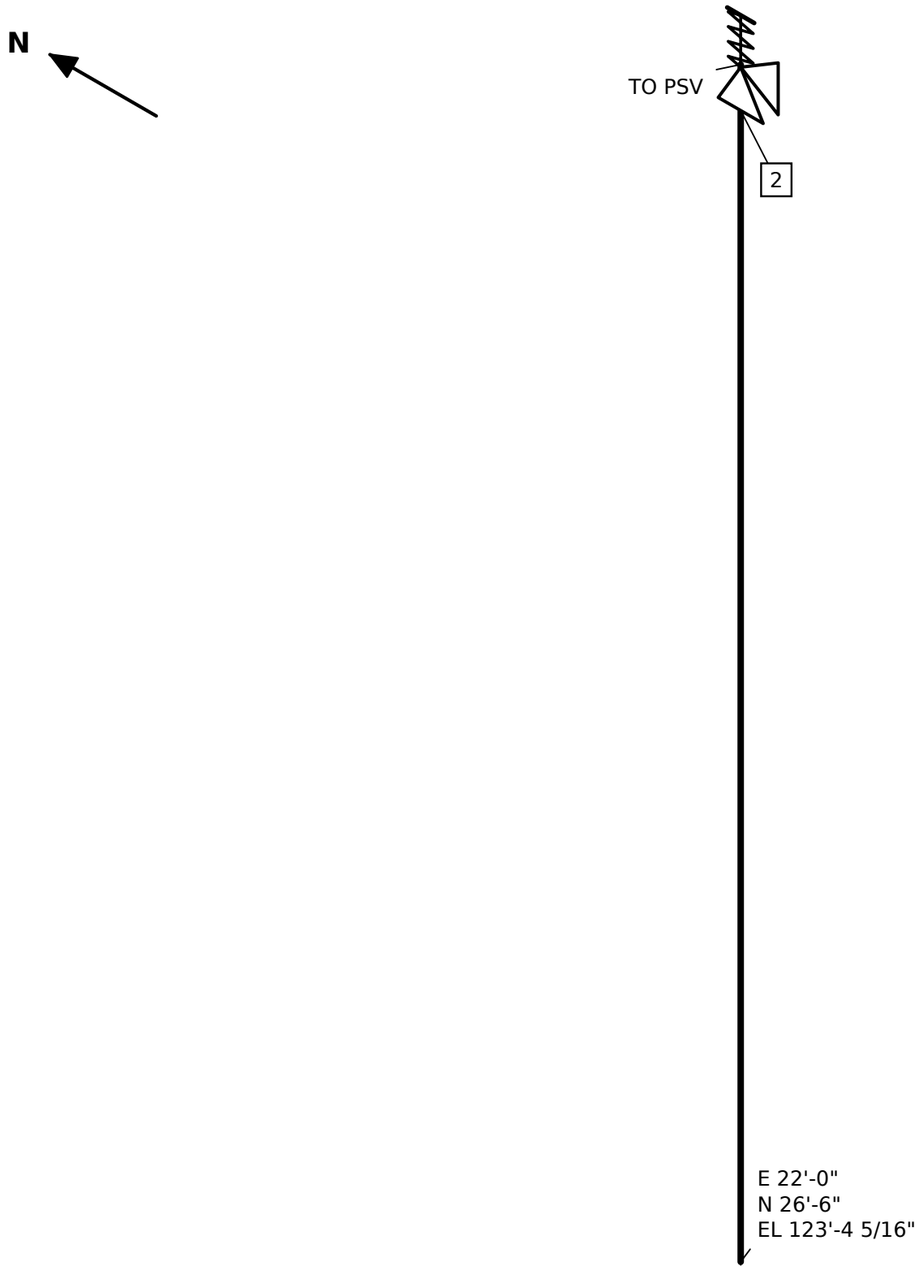
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Relief valve removed; nozzle and outlet blinded	SW-WO-3	Both openings blinded and tagged	Isolation certificate	W	W	—				
3	As-found bench test (pop pressure recorded before any service)	SW-TEST-3	Recorded	Bench test record	W	W	W				
4	Set pressure and seat tightness after service	SW-TEST-3, PSV-1	Set at 150 psi; seat tight	Bench test record, seal tag	H	H	W				
5	Valve reinstalled; new gaskets; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
3"-P150-RV-4002/2	affected	ISO-RV-4002	1	
3"-P150-RV-4002/1	affected	ISO-RV-4002	1	
4"-P150-FL-4005/A1	affected	ISO-FL-4005	1	
3"-P150-RV-4002/1	isolation: blind	ISO-RV-4002	1	
4"-P150-FL-4005/A1	isolation: blind	ISO-FL-4005	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	3"	GASKET 3" P150 COMPRESSED SHEET
2	1	3"x4"	VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET
3	1	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P150

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. HYDROTEST BEFORE PAINTING
4. PAINT PER PROJECT SPECIFICATION

TEST PRESS. -	INSULATION **NONE**	PAINT -	
PWHT -	NDE -	FLUID **RV**	

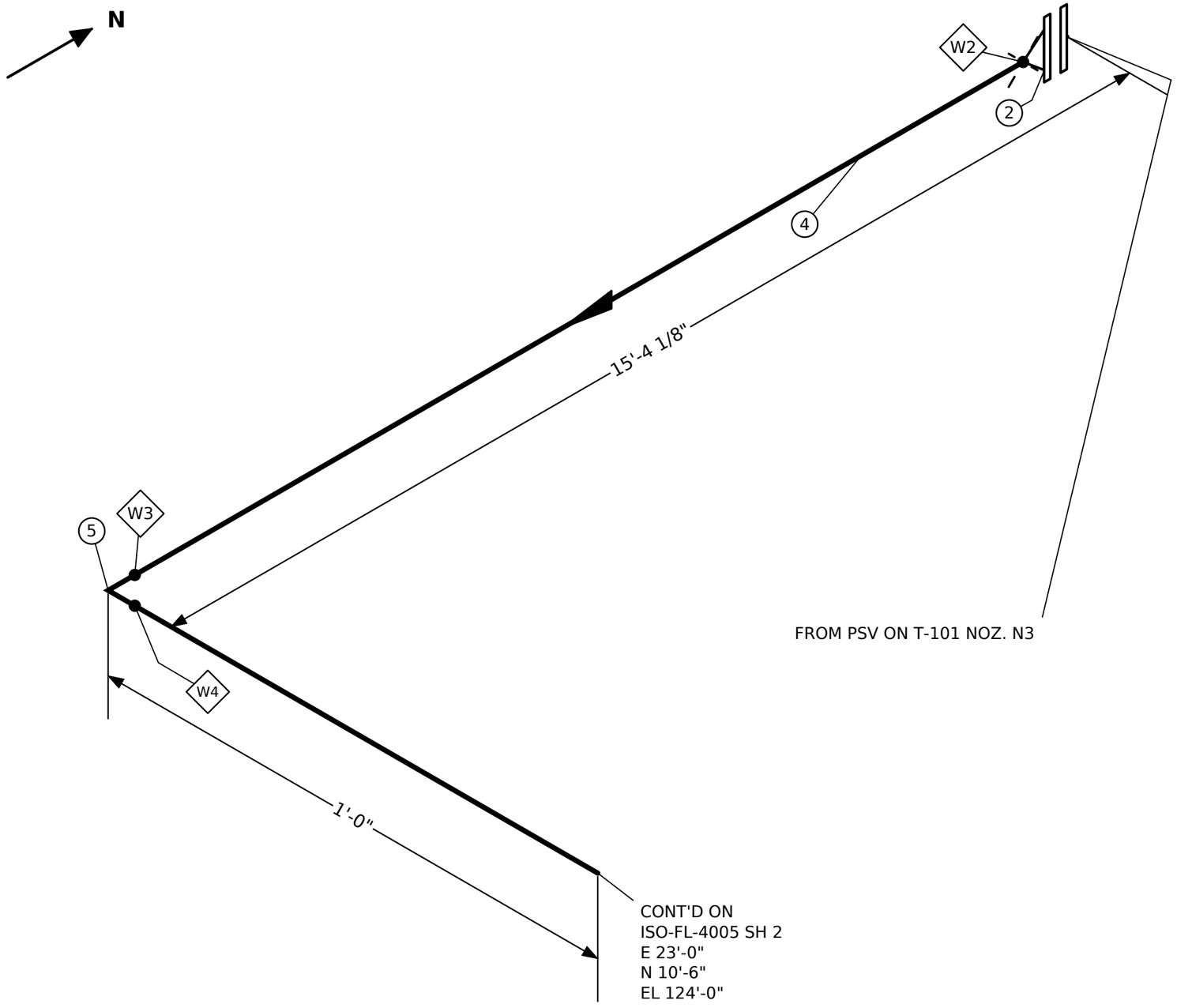
COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

LINE NO. **3"-P150-RV-4002**	PIPING CLASS **H150**	DESIGN PRESS. **150 PSIG**	DESIGN TEMP. **-**
DRAWING NO. **ISO-RV-4002**		SHEET **1 OF 1**	REV **1**
DATE **2026-11-01**	DRAWN **-**	CHECKED **-**	SCALE **NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	4"	GASKET 4" P150 COMPRESSED SHEET
2	1	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
3	1	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
4	14'-9 1/4"	4"	PIPE 4" P150 BEVELLED ENDS
5	1	4"	ELBOW 90° R = 5/3 OD 4" P150

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-
1	2026-11-02	REVISED TO MATCH THE MODEL (FL-1)	-

NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-

PWHT	NDE	FLUID
-	-	FL

COMPANY

SMARTWELD

TITLE

FLARE AND RELIEF OUTLETS

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P150-FL-4005	F150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-FL-4005	1 OF 3	1

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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